StockSense

Adaptive Inventory Replenishment Engine

Project Report — problem, solution, users, AI integration, differentiation and architecture

Executive summary. Businesses lose money in two directions: ordering too late causes stockouts, and ordering too early ties up cash and storage. **StockSense** is a focused, explainable decision-support engine that recommends **when to reorder and how much to order**. It combines demand forecasting, uncertainty-aware safety stock, cost-based order quantities, anomaly handling, forecast evaluation against a baseline, and scenario simulation of demand surges and supplier delays. All inventory mathematics is performed by deterministic, tested code; AI is used only to suggest labels and explain results — always with the formulas, assumptions and user-selected service level visible.

1. Problem Understanding

Every business that holds stock must repeatedly answer two questions: **when** should we place the next order, and **how much** should we order? The answers are hard because demand and supplier behaviour are both uncertain, and the cost of a wrong answer runs in both directions.

Problem	What goes wrong	Why it is hard
Ordering too late	Stockouts and lost sales	Reorder timing must account for supplier lead time and demand during that lead time
Ordering too early	Higher storage cost; cash tied up in inventory	Excess safety margin is invisible until costs accumulate
Unexpected demand increases	Shortages before the next delivery arrives	Demand variability must be quantified, not guessed
Supplier delays	Planned stock arrives after it is needed	Lead times vary; one slow delivery can empty the shelf
Unusual sales data	Forecasts distorted by promotions, errors or stockout-suppressed sales	A spike or a zero can be real demand, a promotion, a data error, or lost demand — and each needs different treatment

Problem statement. Planners need a transparent way to turn sales history and supplier information into a trustworthy reorder point and order quantity — one that cleans unusual data, quantifies its own forecast accuracy, and shows how the recommendation would change if demand or supply conditions change.

Success criteria.

- Recommendations that are reproducible and traceable to explicit formulas and assumptions.
- Forecast accuracy measured on held-out data and compared against a simple baseline.
- Unusual demand handled deliberately rather than silently absorbed into the forecast.
- Stockout and cost risk visible under normal and disrupted conditions.

2. Proposed Solution

StockSense is an **adaptive, explainable replenishment decision-support system** — not merely an inventory dashboard. It ingests sales history and SKU parameters, cleans and forecasts demand, sizes safety stock for a user-selected service level, computes the reorder point and order quantity, estimates cost and risk, and stress-tests the result under disruption scenarios.

2.1 Core inventory calculations

Four established concepts form the foundation: **demand forecasting** (how much will customers need?), **safety stock** (how much extra do we hold?), **reorder point** (when do we reorder?) and **supply planning** (what additional supply is needed, and when?). The engine implements them as follows, where μ_d is average daily demand, σ_d its standard deviation, L the lead time in days, σ_L the lead-time standard deviation and z the factor for the service level the user selects.

$$\text{Reorder Point} = (\mu_d \times L) + \text{Safety Stock}$$

$$\text{Safety Stock} = z \times \sigma_d \times \sqrt{L} \quad \text{(fixed lead time)}$$

$$\text{Safety Stock} = z \times \sqrt{(L \times \sigma_d^{\,2} + \mu_d^{\,2} \times \sigma_L^{\,2})} \quad \text{(variable lead time)}$$

$$\text{EOQ} = \sqrt{(2 \times D \times S \div H)} \quad \text{(D annual demand · S cost per order · H holding cost/unit/year)}$$

Worked example. With daily demand of 10 units, a 5-day lead time and 20 units of safety stock, ROP = (10 × 5) + 20 = **70 units**. If instead σ_d = 4 units and the user selects a 95% service level (z = 1.645), safety stock is 1.645 × 4 × √5 ≈ 14.7, rounded up to 15, giving a reorder point of 65. An order is triggered when the *inventory position* (on-hand plus on-order stock) falls to or below the reorder point.

2.2 Five differentiating capabilities

Capability	Purpose	How it works
Smart replenishment	Calculate reorder points and suitable order quantities	Safety stock from service level and uncertainty; EOQ adjusted for MOQ, pack size and storage limits
Demand-anomaly detection	Identify unusual sales and classify them	Robust statistical flags; classes: promotion, data error, stockout-censored, unknown; user confirms each label
Forecast evaluation	Show whether the model is actually better than a simple baseline	Holdout split; MAE, RMSE, WAPE and bias for baseline and proposed model, reported side by side
Scenario simulation	Show how decisions change under disruption	Seeded Monte Carlo simulation of demand surges and supplier delays; before-and-after comparison
Cost & risk analysis	Estimate cost and stockout risk	Holding + ordering + stockout cost; stockout probability and expected unmet units

2.3 Explainability and honest measurement

Each recommendation displays the **user-selected service level, the statistical factor (z), the formulas with substituted numbers, all assumptions, and any AI-proposed forecast adjustments**. Forecast quality is measured on held-out data; if the proposed model does not beat the baseline, the system says so and uses the baseline. Simulation results are presented as estimates under stated assumptions — StockSense does not promise guaranteed cost savings.

3. Intended Users

User	Context	What they need from StockSense
Inventory / replenishment planner	Reviews reorder decisions across many SKUs daily	A prioritised reorder list, forecasts they can inspect, and the ability to override or approve anomaly labels
Small-business owner / shop manager	Runs a shop without ERP-level planning tools	Plain-language guidance such as "order N units now", with a simple view of why
Procurement / purchasing lead	Manages suppliers and lead-time risk	What-if analysis for supplier delays and a view of which SKUs are exposed
Operations / finance manager	Balances service levels against working capital	Cost trade-offs between holding, ordering and stockouts at different service levels
Hackathon judges / evaluators	Assess the prototype end to end	One complete, reproducible demo with visible calculations and measured forecast accuracy

StockSense can run beside an existing ERP rather than replacing it: it targets teams that want transparency and scenario thinking on top of — or in place of — opaque planning outputs.

4. AI Integration

StockSense deliberately limits where AI is used. The guiding rule is: **AI may suggest and explain; deterministic code decides and calculates.** The system must work fully with AI switched off.

AI task	Role	Guardrails
Suggest anomaly class (promotion / data error / unknown)	Proposes a label with a confidence score and short rationale	Output validated against a JSON schema; falls back to *unknown* on failure; user must confirm; source logged as "ai"
Explain a recommendation in plain language	Narrates the already-computed numbers and formulas	Receives the computed results as input and cannot change them
Propose forecast adjustments (e.g., an upcoming event)	Surfaces context a statistical model cannot see	Displayed as *AI-proposed*, off by default, applied only when the user accepts
Not delegated to AI: safety stock, reorder point, EOQ, costs, simulation, service-level choice	—	Handled exclusively by deterministic, unit-tested code; user-selected settings are never overridden

Why this design. Inventory decisions involve money and customer service, so every number must be reproducible and auditable. AI adds value where judgement over messy, contextual information is needed — interpreting why a sales spike occurred, or turning formulas into language a shop owner can act on — while the human stays in control of approvals.

5. Differentiation from Existing Solutions

Two major ERP systems illustrate how inventory planning is done today. **SAP S/4HANA** supports inventory management, procurement and supply planning using concepts such as reorder points, safety stock and MRP (inventory data → forecasting → reorder point → MRP → replenishment proposal → purchase order). **Oracle NetSuite** is a cloud ERP that links demand forecasting and supply planning, using a Planning Data Repository and Planning Workbench (sales history → demand forecast → planning data → supply planning → replenishment recommendations → order processing).

Aspect	SAP S/4HANA	Oracle NetSuite	StockSense
Main focus	Enterprise operations and inventory planning	Cloud-based business management and supply planning	Focused, explainable replenishment decision support
Key concepts discussed	Reorder point and MRP	Demand forecasting and supply planning	Forecasting + safety stock + ROP + anomaly handling + scenarios
Forecasting	Historical consumption and configured methods	Moving averages, linear regression and other methods	Baseline vs. proposed model, scored on holdout data
Planning approach	Checks inventory against requirements and thresholds	Evaluates expected demand against available and incoming supply	Inventory position vs. uncertainty-aware ROP at a user-chosen service level
Output	Replenishment proposals and other supply elements	Planned orders and supply-planning recommendations	Order recommendation with formulas, assumptions, risk and cost

Both systems support broader capabilities than shown; the table describes the planning concepts examined for this project, not a full product audit.

What StockSense adds. It does not try to recreate an ERP. Its distinctive contribution is making five things explicit and inspectable in one workflow:

- **Anomaly handling as a first-class step** — unusual sales are flagged, classified and reviewed before they shape the forecast.

- **Forecast accountability** — every model is compared to a simple baseline on held-out data, and the comparison is shown.
- **Visible reasoning** — service level, z-factor, formulas, assumptions and AI adjustments appear next to every recommendation.
- **Disruption simulation** — demand surges and supplier delays can be tested before they happen.
- **Bounded AI** — AI assists with labels and explanations but never performs inventory arithmetic.

6. System Architecture

StockSense uses a layered architecture: a React frontend and a single FastAPI (Python) backend that hosts the deterministic analytics core — forecasting, anomaly detection and decision math run in-process — plus a database and an optional AI module.

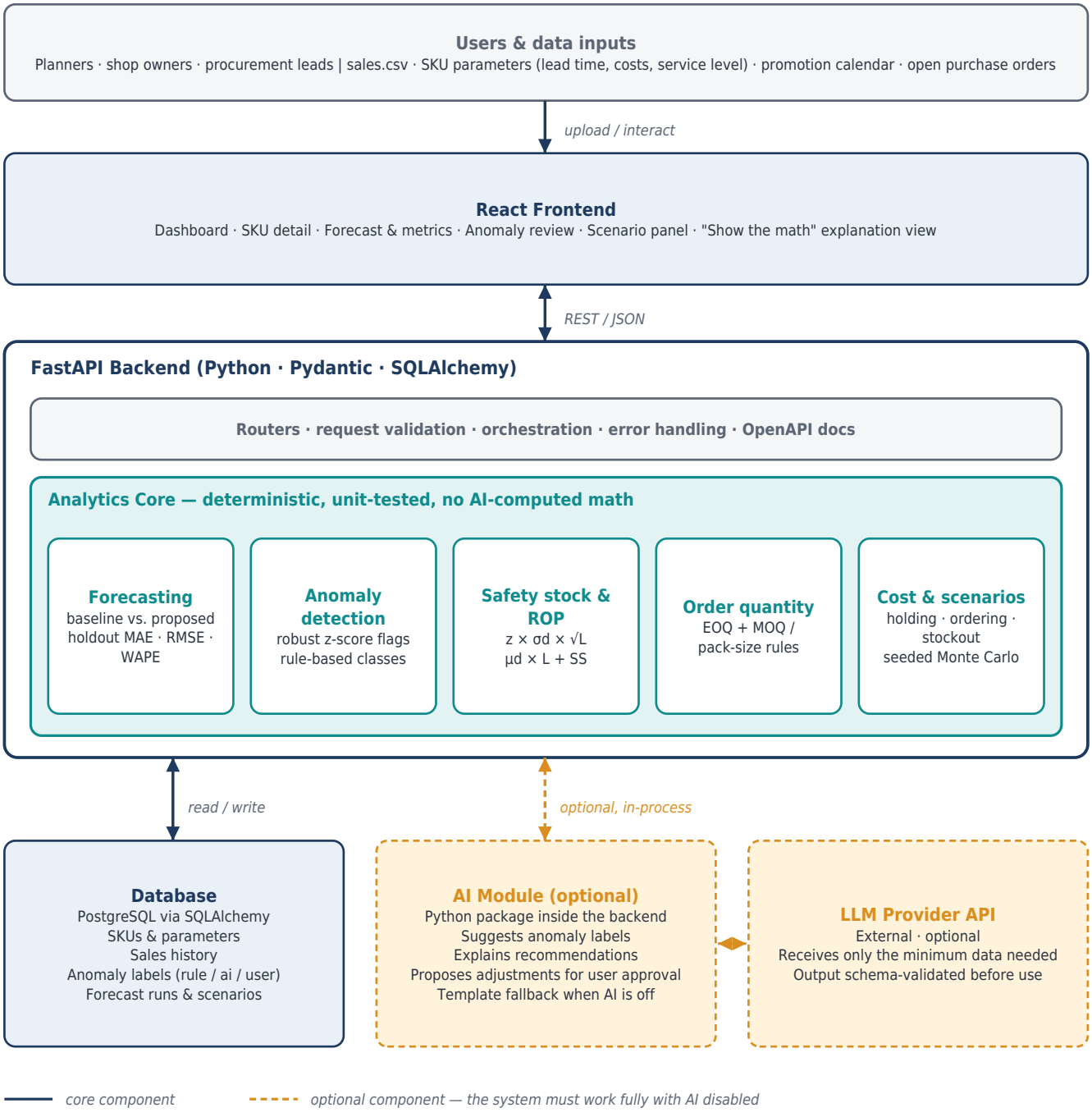


Figure 1 — StockSense component architecture.

6.1 Decision pipeline

Data flows through the stages below. In implementation, anomaly screening runs on raw history before the forecast is fitted (and again on forecast residuals), so cleaned data drives the safety-stock, reorder-point and order-quantity calculations.

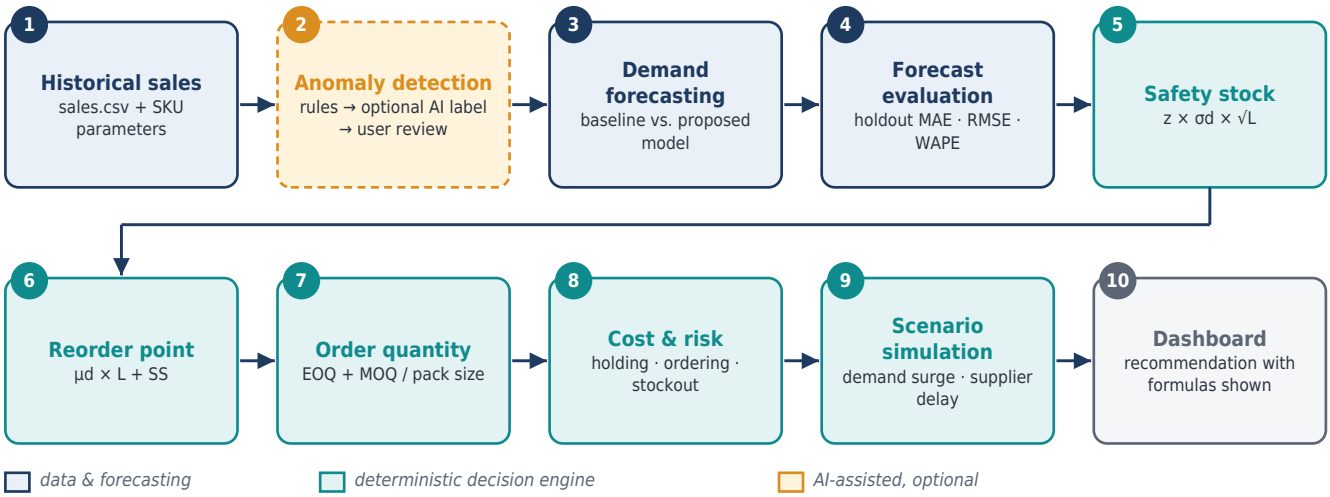


Figure 2 — End-to-end decision pipeline.

6.2 Technology stack

Layer	Choice	Notes
Frontend	React; Recharts or Chart.js	Dashboard, SKU detail, anomaly review, scenario panel, explanation view
Backend	FastAPI + Pydantic v2 + SQLAlchemy	Routers, validation, orchestration, auto-generated OpenAPI documentation
Analytics core	Python: pandas, numpy, scipy, statsmodels	Forecasting, anomaly detection, safety stock, reorder point, EOQ, simulation — no AI-computed math
Database	PostgreSQL	SKUs, sales history, anomalies, adjustments, forecast runs, scenarios
AI	Optional LLM API via a Python module	Anomaly-label suggestions, explanations and proposed adjustments only

6.3 Build plan and validation

The prototype is scoped for an eight-hour build and follows this order: **data contracts → tested decision engine → forecasting and anomaly modules → API integration → frontend → scenario testing and presentation.** Success depends on implementing the core mathematics correctly, agreeing on API contracts early, using realistic data, and delivering one complete end-to-end demo.

> **Evidence and limits.** Formulas are verified with automated tests against worked examples. Forecast performance is reported only from held-out data. Any demonstration data that is synthetic is labelled as such, and the project makes no claim of guaranteed cost savings or forecasting superiority without evidence.